Migration Choices under Informal Insurance.

Eva Pérez Martel
UAB, BSE and CSIC

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Abstract

I study how informal risk-sharing networks shape households' decisions to undertake risky internal migration. The analysis builds on a network model in which households face uncertain rural income, choose whether to migrate at a cost, and share resources with connected households. These links provide insurance, but they can also reduce the private return to migration when migration income is shared. Under CARA preferences, neighbors' migration choices do not affect a household's own decision. Under DARA preferences, however, resources received through the network change households' willingness to bear migration risk. When remittances are shared, a neighbor's migration lowers the migration threshold, generating strategic complementarity. Forming an additional link has an ambiguous effect, reflecting the trade-off between greater insurance and stronger sharing obligations.

Keywords: Risk-sharing, Networks, Migration, Uncertainty

JEL Codes: D81, D85, J61, O12.

1 Introduction

For many rural households in developing countries, migration provides access to economic opportunities that are not available locally. This is closely related to the persistent rural–urban wage gap observed in these economies (Munshi and Rosenzweig, 2016). Yet migration is also a risky investment. A household that sends one of its members to an urban area incurs an upfront cost in exchange for an uncertain future income. For households embedded in informal risk-sharing networks, neither the risks they face nor the returns to migration are necessarily private. Income shocks can be partially absorbed through transfers from other households, while successful migrants may in turn be expected to share part of their earnings through the same relationships. Social connections can therefore provide insurance while at the same time reducing the private return from migration.

This paper asks how a household's position in an informal risk-sharing network affects its willingness to undertake risky migration when migration earnings themselves may be subject to sharing. The answer is not straightforward. Risk-sharing relationships can provide resources that make migration risk easier to bear, but sharing obligations can also reduce the fraction of migration income retained by the migrant. Moreover, when migrants share part of their earnings with connected households, one household's migration decision

changes the resources available to others and may therefore affect their own willingness to migrate.

The main contribution of the paper is to show how risk preferences and the structure of informal sharing relationships jointly shape risky migration decisions. In order to do so, I distinguish between agents that face Constant Absolute Risk Aversion (CARA) and Decresing Absolute Risk Aversion (DARA) preferences. I show that migration decisions are independent under CARA, whereas strategic complementarity arises under DARA.

I study these forces in a network model in which households are connected through an exogenously given set of bilateral risk-sharing relationships. Households face uncertain rural income and decide whether to undertake a costly migration opportunity that generates stochastic remittances. Rural income is shared across existing links, and a fraction of migration income can also be redistributed through the same arrangement. Migration decisions are characterized by a household-specific threshold for the probability of obtaining the high migration return. Network position affects this threshold through two main channels: the resources received through risk sharing and the fraction of migration income that the household retains.

The importance of these channels depends critically on risk preferences. Under CARA, changes in background resources do not affect the household's willingness to bear migration risk. When remittances are entirely private, households therefore face the same migration threshold regardless of their position in the network. When remittances are partially shared, network position matters because more connected households retain a smaller fraction of their migration income and consequently face a higher migration threshold. However, the resources received from other households still factor out of the migration comparison, so neighbors' migration choices do not generate strategic interaction.

The result changes under DARA preferences. In this case, willingness to bear risk depends on background consumption. When remittances are shared, a neighbor's migration increases the household's available resources and makes its own migration opportunity more attractive. As a result, the migration threshold falls. Migration decisions are therefore strategic complements along existing risk-sharing links: migration by one household increases the incentives of connected households to migrate as well.

The model also shows why greater connectivity does not necessarily encourage migration. Forming an additional link gives the household access to resources from a new neighbor, but it also changes how much of its own income it retains within the sharing arrangement. When migration income is partially shared, the new link additionally reduces the fraction of migration income retained by the household. The overall effect of a new connection is therefore generally ambiguous. A new link may encourage migration when the additional resources it provides dominate the associated sharing obligations, but it may discourage migration when the opposite is true.

The paper brings these strands together by studying a risky migration decision within an exogenously given network of bilateral risk-sharing relationships where households have distinct attitudes towards risk. Hence, this paper characterizes how risk preferences, local

network structure, and sharing obligations jointly shape individual migration incentives and equilibrium migration patterns.

Related Literature

This paper relates most closely to the literature studying the interaction between migration and informal insurance. Munshi and Rosenzweig (2016) show that access to caste-based insurance can discourage rural–urban migration because households trade off the income gains from migration against the consumption-smoothing benefits provided by their origin community. Morten (2019) studies the joint determination of temporary migration and endogenous risk sharing, showing that migration can provide a form of self-insurance and reduce the need for informal insurance, while informal insurance can itself reduce the need to migrate. Similarly, Meghir et al. (2022) study the interaction between migration opportunities and informal insurance. They show that temporary migration subsidies can improve risk sharing when migration is risky, whereas a permanent reduction in migration costs can instead reduce informal risk sharing. Relative to these papers, I take the risk-sharing network as given and examine how network position and risk preferences (CARA and DARA) shape migration incentives.

The paper also relates to the literature on risk sharing and risk taking in social networks. Bramoullé and Kranton (2007) study the formation of risk-sharing networks and show that network structure can place otherwise identical individuals in different positions with different insurance outcomes. Bloch et al. (2008) study bilateral informal insurance arrangements sustained through social networks, emphasizing the role of network structure in the sustainability of informal insurance. Empirically, Fafchamps and Gubert (2007) show that risk-sharing relationships in rural Philippines are strongly associated with social and geographical proximity and predict subsequent gifts and loans between households. Particularly related to the mechanism studied here, Belhaj and Deroıan (2012) analyze risk-taking incentives when revenues are partially shared through an exogenous social network. They show that network-based sharing generates heterogeneous risk-taking incentives and, when risky technologies are positively correlated, strategic substitutability in risk choices. In contrast, the strategic interaction in the present paper operates through background resources: when migration income is shared, a neighbor's migration increases the resources available to the household and, under DARA, increases its willingness to undertake its own migration risk.

A related literature studies the social obligations associated with family and kinship networks. Jakiela and Ozier (2016) provide experimental evidence that individuals may sacrifice expected income in order to conceal resources from relatives and neighbors, consistent with social pressure to share observable income. This provides empirical motivation for modeling sharing obligations as an implicit tax on individually generated income. In the present model, these obligations apply to migration income: a fraction of remittances is redistributed through existing risk-sharing relationships, reducing the fraction of the

migration return privately retained by the migrant.

Finally, the paper relates to the literature studying how background resources and background risk affect willingness to undertake risky prospects (Gollier and Pratt, 1996; Gollier, 2001). This channel motivates the distinction between CARA and DARA preferences in the model. Under CARA preferences, background resources factor out of the migration decision. Under DARA preferences, instead, willingness to bear migration risk depends on background consumption. Consequently, resources obtained through the risk-sharing network can affect a household's willingness to undertake its own risky migration opportunity.

The paper proceeds as follows. Section 2 presents the main model. Section 3 provides results, first under CARA preferences and then under DARA preferences. Section 4 concludes with the main contributions of the paper and future extensions.

2 Model

2.1 Primitives

I consider an economy composed of a finite number of households connected through an exogenously given risk-sharing network. Formally, an economy is characterized by the tuple

$$\mathcal{E} = \left(N, G, \Omega, \Pi, e^H, e^L, q, \alpha, (u_i)_{i \in N}\right),$$

where $N = \{1, \ldots, n\}$, with $n < \infty$, denotes the set of households, G the network of bilateral risk-sharing relationships, Ω the state space with probability measure Π, (e^H, e^L) denote the possible remittance realizations, q is the migration cost, α determines the remittance-sharing rule, and u_i is household i's utility function.

Households are represented by nodes in a network and links correspond to bilateral informal risk-sharing relationships through which resources can be transferred across households. I assume that the network is exogenous and fixed throughout the analysis and is represented by the adjacency matrix G that represents the connections between any pair of households.

A link exists between two households i and j whenever $G(i, j) = 1$. The network is undirected so that $G(i, j) = G(j, i)$, and self-loops are excluded from G, so $G(i, i) = 0$. For every household i, I define its neighborhood as $N_i = \{j \in N : G(i, j) = 1\}$, and its closed neighborhood as $\bar{N}_i = N_i \cup \{i\}$, and its degree as $d_i = |N_i|$ that measures the number of direct risk-sharing partners of household i, while $\bar{d}_i = |\bar{N}_i|$.

I model uncertainty through a finite state space $\Omega = \{\omega_1, \ldots, \omega_K\}$[1], where each state specifies one realization of household income and migration remittances. The probability assigned to each state is given by the probability mass function $\Pi : \Omega \to [0, 1]$.

Households ex ante commit to bilateral state-contingent transfers, which determine the risk-sharing arrangement represented by the network given the migration profile. The

[1]Since Ω is finite, it is equipped with the power-set sigma algebra $\mathcal{F} = 2^{\Omega}$, which contains all subsets of Ω, including $\emptyset$ and Ω itself.

transfer rule is a collection of functions $t_{ij} : \Omega \times \{0,1\}^n \to \mathbb{R}$ one for each linked pair of households. Whenever a transfer relationship exists between households i and j, transfers satisfy the following $t_{ij}(\omega, \mathbf{M}) = -t_{ji}(\omega, \mathbf{M})$, for every state $\omega \in \Omega$ and vector of migration choices M. Throughout the paper, the existence of these bilateral transfers will be referred to as informal risk-sharing or kinship taxation.

Household income is stochastic that $y_i(\omega) : \Omega \to \mathbb{R}$ denote the endowment (i.e. rural or background income) of household i in state ω. It follows a binary stochastic process $y^H = W + \tilde{y}^H$ and $y^L = W + \tilde{y}^L$, where $W > 0$ is given and exogenous initial wealth level. It occurs with probabilities ξ and $1 - \xi$, respectively, and $\tilde{y}^L < 0 < \tilde{y}^H$.

Each household chooses whether to migrate, $M_i \in \{0,1\}$, where migration is interpreted as a risky investment opportunity. This opportunity is interpreted as sending a member of the household to an urban economy. Migration generates stochastic remittances $e_i(\omega) : \Omega \to \mathbb{R}$, and requires paying an exogenous migration cost $q > 0$. Throughout the analysis I assume two possible remittance realizations e^H and e^L, which occur with probabilities p_i and $1 - p_i$, respectively. I allow the probability of receiving the high migration return to be household-specific, p_i. This captures heterogeneity across households in their migration prospects. This formulation is related to Bryan et al. (2014), who model households as uncertain about whether they are "good at migrating", interpreting this uncertainty either as an unknown household-specific suitability for migration or as the probability of finding a useful connection at the destination. I then allow the corresponding prior probability to vary across households.

Migration income is only partially shared through the informal insurance arrangement, and this captures the intensity of kinship taxation. A fraction $\alpha \in (0,1)$ of remittances is transferred according to the risk-sharing agreement, while the remaining fraction $(1 - \alpha)$ that is is privately retained by the migrating household.

Transfers are determined by an exogenous sharing rule that depends on households' network positions. In particular,

$$t_{ij}(\omega, \mathbf{M}) \equiv t_{ij}(\omega, M_i, M_j) = \frac{y_i(\omega) + \alpha e_i(\omega) M_i}{\bar{d}_i} - \frac{y_j(\omega) + \alpha e_j(\omega) M_j}{\bar{d}_j}$$

so that both rural or background income and the transferable component of remittances are redistributed across connected households according to their degrees. Notice that migration income depends on the migration choices of household i and its peers j.

Rural income is always subject to kinship taxation: the informal obligation to share resources with connected households in the kinship or risk-sharing network. Such transfers provide informal insurance against idiosyncratic income risk, but at the same time imply that households do not fully retain their realized income. This reduced-form sharing rule captures the informal risk-sharing arrangements documented in village economies (Townsend, 1994; De Weerdt and Dercon, 2006), as well as the social pressure to share resources with kin and other members of the social network, sometimes referred to as "forced solidarity" or an implicit "kinship tax" (Jakiela and Ozier, 2016). Hence, households

will face informal commitment and will always share resources with their neighbors in order to smooth consumption. This happens even when no migration takes place, so each household's rural income will be distributed across peers through this self-enforced insurance. Household consumption in state ω is therefore given by

$$c_i(\omega, \mathbf{M}) = y_i(\omega) - \sum_{j \in N_i} t_{ij}(\omega, M_i, M_j) + (e_i(\omega) - q)M_i. \tag{1}$$

Preferences are represented by a utility function over consumption as follows $u_i(c_i)$, where $u_i(\cdot)$ is assumed to be increasing, strictly concave and twice continuously differentiable.

To rule out trivial results where a household will not migrate, even the more connected household needs to obtain positive migration gain under the H state. I assume that *Assumption* 2.1.

$$\min_{i \in N} \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^H - q > 0$$

$\forall \bar{d}_i$.

2.2 Timing and the Household Problem

The sequence of events in this economy is as follows. The exogenous risk-sharing network G is given, and households commit ex ante to the bilateral transfer arrangement implied by the sharing rule. Nature then draws the vector of income and migration shocks, determining a state $\omega \in \Omega$. Households know the probability distribution governing these shocks but do not observe their realizations when making their migration decisions. They simultaneously choose whether to migrate, $M_i \in \{0, 1\}$, taking the network structure and the sharing rule as given. Income, migration outcomes, and bilateral transfers are then realized according to the state ω, determining household consumption. Finally, consumption takes place and households obtain utility.

Each household chooses its migration decision to maximize expected utility:

$$\max_{M_i \in \{0,1\}} \mathbb{E}\left[u_i(c_i(\omega, \mathbf{M})\right]$$

where the expectation is taken over the compound probability distribution for endowment (rural income) and migration income and consumption c_i is given as in Equation 1.

2.3 Shocks and Equilibrium

The economy is subject to two sources of uncertainty: household income and migration returns. Income realizations are exogenous and stochastic. Each household's income can take one of two possible values, $y_i(\omega) \in \{y^H, y^L\}$. The high-income realization occurs with probability ξ, while the low-income realization occurs with probability $1 - \xi$.

Conditional on choosing migration ($M_i = 1$), household i receives stochastic remittances.

Migration income follows a binary lottery, $e_i(\omega) \in \{e^H, e^L\}$ where the high-remittance realization e^H occurs with probability p_i, and the low-remittance realization $e^L > 0$ occurs with probability $1 - p_i$. Throughout the analysis I assume $e^H - q > 0$, and $e^L - q < 0$, so that migration generates positive net gains in the good state and losses in the bad state.

The state space is $\Omega = \{y^H, y^L\}^N \times \{e^H, e^L\}^N$. A state $\omega = (\mathbf{y}, \mathbf{e}) \in \Omega$ and it specifies the realization of both shocks for every household, where $\mathbf{y} = (y_1, \ldots, y_n)$ and $\mathbf{e} = (e_1, \ldots, e_n)$. I distinguish between the state space and the income realizations that are relevant for a particular household. For each household i, define the local income state space

$$\Omega_i^y \equiv \{y^H, y^L\}^{|\bar{N}_i|}.$$

An element $\omega_i^y \in \Omega_i^y$ specifies the realization of rural income for household i and all households in its closed neighborhood $\bar{N}_i$. The corresponding probability function is

$$\Pi_i^y(\omega_i^y) \equiv \prod_{k \in \bar{N}_i} \left[\xi^{1\{y_k(\omega_i^y) = y^H\}} (1 - \xi)^{1\{y_k(\omega_i^y) = y^L\}} \right].$$

Hence, $\Pi : \Omega \to [0, 1]$ is given by

$$\Pi(\omega) = \prod_{i \in N} \left[\xi^{1\{y_i = y^H\}} (1 - \xi)^{1\{y_i = y^L\}} p_i^{1\{e_i = e^H\}} (1 - p_i)^{1\{e_i = e^L\}} \right]$$

where I is the indicator function of the household getting a particular realization for income.

Finally, a migration profile $M^* = (M_1^*, \ldots, M_n^*)$ is a Nash equilibrium if, for every household $i \in N$:

$$\mathbb{E}\{u_i(M_i^*, M_{-i}^*)\} \geq \mathbb{E}\{u_i(M_i, M_{-i}^*)\}, \qquad \forall\, M_i \in \{0, 1\}.$$

3 Results

3.1 CARA Preferences

I first characterize the equilibrium of the migration game under CARA preferences. Throughout this section, households have identical preferences represented by $u(c) = -e^{-rc}$, where $r > 0$ denotes the coefficient of absolute risk aversion. I begin with the benchmark case without kinship taxation, then introduce partial remittance sharing, and finally provide comparative statics.

3.1.1 Non-Kinship Taxation

Consider first the benchmark economy in which migration income is entirely private ($\alpha = 0$). In this case, households do not share remittances through the informal insurance network. Consequently, migration only affects the consumption of the household undertaking the

migration decision. Although households remain connected through the network and may share labor income according to the exogenous transfer rule, migration income is privately retained and is therefore not subject to kinship taxation.

Lemma 3.1 (Migration Threshold under Private Remittances)**.** *In the absence of kinship taxation, household i chooses to migrate if and only if $p_i \geq \phi$, where*

$$\phi = \frac{e^{-r(e^L - q)} - 1}{e^{-r(e^L - q)} - e^{-r(e^H - q)}}.$$

The migration threshold is identical across households and depends only on the parameters of the migration lottery and the degree of absolute risk aversion and thus, under CARA preferences, migration decisions are independent of background wealth. Therefore, conditional on the network structure, household i chooses whether to migrate by comparing the expected utility associated with migration with the utility obtained from remaining in the village. Consequently, the migration decision depends exclusively on the expected gains and losses generated by the migration lottery and the risk-sharing arrangement.

To rule out any generic multiplicity in equilibrium by households being indifferent between migrating or not, I assume

Assumption 3.1. For every household $i \in N$ and every migration profile M_{-i}, $p_i \neq \phi_i(M_{-i}; \mathbf{p}_{-i})$

Proposition 3.1 (Equilibrium under Private Remittances)**.** *A migration profile $M^* = (M_1^*, \ldots, M_n^*)$ is a Nash equilibrium if, for every household $i \in N$,*

$$M_i^* = \begin{cases} 1, & \text{if } p_i > \phi, \\ 0, & \text{if } p_i < \phi. \end{cases}$$

Here the equilibrium is unique and households face identical ϕ.

The absence of remittance sharing also eliminates strategic interaction in migration choices. A household's optimal migration decision does not depend on whether its neighbors migrate, so each household can determine its action independently. Away from indifference, the equilibrium is therefore unique. Heterogeneity in migration choices can still arise when households have different probabilities p_i of obtaining the high migration return, but this heterogeneity does not reflect interaction between their decisions.

3.1.2 Partial Kinship Taxation

I now extend the benchmark economy by allowing households to share part of their migration income through the informal insurance network. In particular, a fraction $\alpha \in (0, 1)$ of remittances is transferred according to the exogenous sharing rule, while the remaining fraction $(1 - \alpha)$ that is privately retained by the migrating household. Consequently, migration income becomes partially subject to kinship taxation.

I show that unlike the benchmark case, migration decisions now depend on the house-

hold's position in the network. More connected households must distribute the transferable component of remittances among a larger number of neighbors, reducing the private return from migration.

Lemma 3.2 (Migration Threshold under Kinship Taxation)**.** *Household i chooses to migrate if and only if $p_i \geq \phi_i$ where the household-specific threshold probability for migration is*

$$\phi_i = \frac{e^{-r\left(\alpha\frac{e^L}{\bar{d}_i}+(1-\alpha)e^L-q\right)}-1}{e^{-r\left(\alpha\frac{e^L}{\bar{d}_i}+(1-\alpha)e^L-q\right)}-e^{-r\left(\alpha\frac{e^H}{\bar{d}_i}+(1-\alpha)e^H-q\right)}}$$

Once migration income is partially shared, the migration threshold becomes household-specific even under CARA preferences. The reason is that household i retains only the fraction $1-\alpha+\frac{\alpha}{\bar{d}_i}$ of its own migration income. A more connected household therefore keeps a smaller fraction of both the high and low remittance realizations. Network position now affects the private return to migration even though background resources continue to factor out of the migration decision under CARA preferences.

The following proposition characterizes the migration threshold under partial kinship taxation.

Proposition 3.2 (Equilibrium under Partial Kinship Taxation)**.** *Suppose households have CARA preferences and remittances are partially subject to kinship taxation. Let ϕ_i denote household i's migration threshold. Then a migration profile $M^* = (M_1^*, \ldots, M_n^*)$ is a Nash equilibrium if and only if, for every household $i \in N$,*

$$M_i^* = \begin{cases} 1, & \text{if } p_i > \phi_i, \\ 0, & \text{if } p_i < \phi_i, \end{cases}$$

Observe that partial kinship taxation generates heterogeneity in migration incentives across network positions, but it does not by itself generate strategic interaction under CARA preferences. Each household's threshold depends on its own degree, but not on the contemporaneous migration choices of its neighbors. The equilibrium can be heterogeneous because households with different degrees or different p_i can lie on different sides of their respective migration thresholds.

3.1.3 Comparative Statics

For ease of exposition, define household i's net migration returns in the high and low states as

$$H_i = \left((1-\alpha)+\frac{\alpha}{\bar{d}_i}\right)e^H - q, \qquad L_i = \left((1-\alpha)+\frac{\alpha}{\bar{d}_i}\right)e^L - q,$$

The comparative statics summarize how the characteristics of the migration lottery and the sharing arrangement affect the probability of success required for migration. Better migration outcomes make migration more attractive: a larger high-state return or a less

severe low-state loss lowers the threshold, whereas a larger migration cost or greater risk aversion raises it. Under CARA preferences, changes in background income have no effect on migration incentives. In contrast, when $\alpha > 0$, greater connectivity raises the migration threshold because it reduces the fraction of the household's own migration income that is privately retained.

Parameter	**Effect on ϕ_i**	**Interpretation**
High-state return H_i	$\frac{\partial \phi_i}{\partial H_i} < 0$	A larger successful-state return makes migration more attractive and lowers the probability of success required for migration.
Low-state return L_i	$\frac{\partial \phi_i}{\partial L_i} < 0$	A larger low-state return makes an unsuccessful migration outcome less costly and lowers the migration threshold.
Migration cost q	$\frac{\partial \phi_i}{\partial q} > 0$	A higher migration cost reduces net migration returns in both states and raises the probability of success required for migration.
Risk aversion r	$\frac{\partial \phi_i}{\partial r} > 0$	More risk-averse households require a higher probability of receiving the high migration return.
Background income $y(\omega)$	$\frac{\partial \phi_i}{\partial y_i(\omega)} = 0$	Under CARA preferences and independence between migration and background income, background income does not affect the migration threshold.
Retained fraction $\frac{1}{\bar{d}_i}$	$\frac{\partial \phi_i}{\partial (1/\bar{d}_i)} < 0 \quad \text{if } \alpha > 0$	When remittances are shared, a larger retained fraction increases migration returns and lowers the migration threshold.
Degree $\bar{d}_i$	$\frac{\partial \phi_i}{\partial \bar{d}_i} > 0 \quad \text{if } \alpha > 0$	A larger sharing network reduces the fraction of remittances retained by household i and therefore raises the migration threshold.
Degree without remittance sharing	$\frac{\partial \phi_i}{\partial (1/\bar{d}_i)} = \frac{\partial \phi_i}{\partial \bar{d}_i} = 0 \quad \text{if} \quad \alpha = 0$	When migration returns are not distributed across peers, degree does not affect the migration threshold.

Table 1: Comparative statics of the migration threshold.

3.2 DARA Preferences

I now characterize the equilibrium of the migration game under DARA preferences. Throughout this section, I assume that households have identical DARA preferences with strictly concave $u(c_i)$.

After substituting the transfer rule into the budget constraint, household i's consumption

is

$$c_i(\omega, \mathbf{M}) = \frac{y_i(\omega)}{\bar{d}_i} + \sum_{j \in N_i} \frac{y_j(\omega)}{\bar{d}_j} + \alpha \sum_{j \in N_i} \frac{e_j(\omega) M_j}{\bar{d}_j} + \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e_i(\omega) M_i - q M_i.$$

To guarantee positive consumption, I assume

Assumption 3.2 (Positive consumption). For every household $i \in N$,

$$y^L \left(\frac{1}{\bar{d}_i} + \sum_{j \in N_i} \frac{1}{\bar{d}_j} \right) + \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^L - q > 0.$$

Assume that remittances are positive, although the low remittance realization does not cover the migration cost $0 < e^L < q < e^H$.

The condition is evaluated at the lowest possible consumption realization for household i. This occurs when household i and all its neighbors receive the low background-income realization, household i migrates and receives e^L, and none of its neighbors migrates. Indeed, since $\alpha > 0$ and $e^L > 0$, migration by any neighbor j would generate the additional positive contribution $\alpha e_j / \bar{d}_j$ to household i's consumption. Hence, consumption is minimized when $M_j = 0$ for every $j \in N_i$. In contrast, since $e^L < q$ and $1 - \alpha + \frac{\alpha}{\bar{d}_i} \leq 1$, household i's own low migration realization decreases its consumption relative to not migrating.

Since $y^L = W + \tilde{y}^L$, Assumption 3.2 can equivalently be written as

$$W > \max_{i \in N} \left(\frac{q - \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^L}{\frac{1}{\bar{d}_i} + \sum_{j \in N_i} \frac{1}{\bar{d}_j}} - \tilde{y}^L \right). \tag{2}$$

When migration income is not subject to kinship taxation, $\alpha = 0$, the condition reduces to

$$W > \max_{i \in N} \left(\frac{q - e^L}{\frac{1}{\bar{d}_i} + \sum_{j \in N_i} \frac{1}{\bar{d}_j}} - \tilde{y}^L \right). \tag{3}$$

The positive-consumption restriction is therefore weakly more restrictive when migration income is subject to kinship taxation (Equation 2). In particular, $1 - \alpha + \frac{\alpha}{\bar{d}_i} \leq 1$ implies $q - \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^L \geq q - e^L$.

Thus, for a given network and background-income distribution, a weakly larger value of W is required to guarantee positive consumption under kinship taxation. The inequality is strict whenever $\alpha > 0$ and $\bar{d}_i > 1$ for the household determining the maximum.

3.2.1 Non-Kinship Taxation

I consider first the benchmark economy in which migration income is entirely private ($\alpha = 0$), and not subject to kinship taxation.

Assumption 3.3 (No Indifference). For every household $i \in N$ and every migration profile M_{-i}, $p_i \neq \phi_i(M_{-i}; \mathbf{p}_{-i})$.

Lemma 3.3 (Equilibrium Characterization under DARA)**.** *Under DARA preferences and no kinship taxation on remittances, household i chooses to migrate if and only if $p_i \geq \phi_i$ where the household-specific threshold probability for migration is*

$$\phi_i = \frac{\sum_{\omega_i^y} \Pi_i^y(\omega_i^y) \left[u(Y_i(\omega_i^y)) - u(Y_i(\omega_i^y) + e^L - q)\right]}{\sum_{\omega_i^y} \Pi_i^y(\omega_i^y) \left[u(Y_i(\omega_i^y) + e^H - q) - u(Y_i(\omega_i^y) + e^L - q)\right]},$$

where $Y_i(\omega_i^y) = \frac{y_i(\omega)}{\bar{d}_i} + \sum_{j \in N_i} \frac{y_j(\omega)}{\bar{d}_j}$.

The threshold ϕ_i depends on the distribution of household i's background resources and thus on its network position, but not on the migration choices or migration probabilities of its neighbors.

Proposition 3.3 (Equilibrium under Non-kinship Taxation)**.** *A migration profile $M^* = (M_1^*, \ldots, M_n^*)$ is a Nash equilibrium if, for every household $i \in N$,*

$$M_i^* = \begin{cases} 1, & \text{if } p_i > \phi_i, \\ 0, & \text{if } p_i < \phi_i. \end{cases}$$

Moreover, the Nash Equilibrium is unique.

The proposition implies that the equilibrium may be homogeneous or heterogeneous depending on the household-specific comparisons between p_i and ϕ_i. Although network position affects the migration threshold under DARA preferences, private remittances imply that one household's migration does not change the resources of its neighbors. There is therefore no contemporaneous strategic interaction in migration choices when $\alpha = 0$. Notice that each household has a uniquely determined migration decision and the Nash equilibrium is unique, although it may contain both migrating and non-migrating households because their thresholds differ.

3.2.2 Partial Kinship Taxation

I now extend the benchmark economy by allowing households to share part of their migration income through the informal insurance network.

Proposition 3.4 (Equilibrium Characterization under DARA and Partial Kinship Taxation)**.** *Household-specific threshold probability for migration is $\phi_i(M_{-i}; \mathbf{p}_{-i}) \in (0, 1)$ given*

by

$$\phi_i(M_{-i};\mathbf{p}_{-i}) = \frac{\sum_{\omega^y}\Gamma_i(\omega;M_{-i},\mathbf{p}_{-i})\left[u(Y_i(\omega,M_{-i})) - u(Y_i(\omega,M_{-i})+L_i)\right]}{\sum_{\omega^y}\Gamma_i(\omega;M_{-i},\mathbf{p}_{-i})\left[u(Y_i(\omega,M_{-i})+H_i) - u(Y_i(\omega,M_{-i})+L_i)\right]},$$

where

$$Y_i(\omega,M_{-i}) = \frac{y_i(\omega)}{\bar{d}_i} + \sum_{j\in N_i}\left[\frac{y_j(\omega)}{\bar{d}_j} + \frac{\alpha e_j(\omega)M_j}{\bar{d}_j}\right],$$

$$\Gamma_i(\omega;M_{-i},\mathbf{p}_{-i}) \equiv \Pi_i^y(\omega_i^y)\prod_{j\in N_i,M_j=1}\left[p_j^{1e_j(\omega)=e^H}(1-p_j)^{1e_j(\omega)=e^L}\right]$$

and

$$H_i = \left(1-\alpha+\frac{\alpha}{\bar{d}_i}\right)e^H - q, \qquad L_i = \left(1-\alpha+\frac{\alpha}{\bar{d}_i}\right)e^L - q.$$

Moreover, a migration profile $M^* = (M_1^*,\ldots,M_n^*)$ is a Nash equilibrium if, for every household $i \in N$,

$$M_i^* = \begin{cases} 1, & \text{if } p_i > \phi_i(M_{-i}^*;\mathbf{p}_{-i}), \\ 0, & \text{if } p_i < \phi_i(M_{-i}^*;\mathbf{p}_{-i}). \end{cases}$$

With DARA utility and partially shared remittances, the migration threshold depends both on household i's own network position and on the migration choices of its neighbors. When a neighbor migrates, part of its remittance income enters household i's background resources through the sharing arrangement. Because willingness to bear risk depends on background consumption under DARA preferences, these additional resources can change household i's own migration incentives. This is the channel through which migration choices become linked across households.

Proposition 3.5 (Strategic Complementarity under Partial Kinship Taxation)**.** *Household i's migration decision exhibits strategic complementarity with that of any neighbor $j \in N_i$:*

$$\phi_i(M_j = 1, M_{-i,-j};\mathbf{p}_{-i}) < \phi_i(M_j = 0, M_{-i,-j};\mathbf{p}_{-i}).$$

The proposition implies that a neighbor's migration lowers household i's migration threshold, so migration decisions are strategic complements along existing risk-sharing links. Then conditional on the migration choices of all other households (excluding i and j), an increase in neighbor j's migration decision lowers household i's migration threshold. When household j migrates, it contributes positive remittance income to household i through the sharing arrangement. This raises household i's background consumption and, under decreasing absolute risk aversion, makes the risky migration opportunity relatively more attractive. Consequently, household i requires a lower probability of receiving the high

migration return before choosing to migrate. Thus, a neighbor's migration increases the incentives for migration rather than crowding them out.

3.2.3 Comparative Statics

I now study how the migration threshold changes when household i forms an additional link with household $j \notin N_i$. Since degree is discrete, I compare the original network G with the network $G + ij$. Throughout this subsection, I fix M_{-i} and $\mathbf{p}_{-i}$ and suppress these arguments when this does not create confusion.

Recall that, under network G, household i's resources excluding its own migration return are

$$Y_i^G(\omega) = \frac{y_i(\omega)}{\bar{d}_i} + \sum_{k \in N_i} \left[\frac{y_k(\omega)}{\bar{d}_k} + \alpha \frac{e_k(\omega) M_k}{\bar{d}_k} \right].$$

After household i forms the additional link with $j \notin N_i$, these resources become

$$Y_i^{G+ij}(\omega) = \frac{y_i(\omega)}{\bar{d}_i + 1} + \sum_{k \in N_i} \left[\frac{y_k(\omega)}{\bar{d}_k} + \alpha \frac{e_k(\omega) M_k}{\bar{d}_k} \right] + \frac{y_j(\omega) + \alpha e_j(\omega) M_j}{\bar{d}_j + 1}.$$

Proposition 3.6 (Effect of an Additional Link on the Migration Threshold)**.** *Suppose household i forms a link with household $j \notin N_i$, the additional link changes household i's resources excluding its own migration return by*

$$Y_i^{G+ij}(\omega) - Y_i^G(\omega) = \frac{y_j(\omega) + \alpha e_j(\omega) M_j}{\bar{d}_j + 1} - \frac{y_i(\omega)}{\bar{d}_i(\bar{d}_i + 1)}.$$

The effect of the link on household i's background resources is generally ambiguous: forming an additional risk-sharing link has two competing effects on household i's migration incentives. It provides access to the new neighbor's rural income and potential remittances, but also reduces the fraction of the household's own rural income that it retains. The resulting change in background resources can therefore be positive or negative. Moreover, when $\alpha > 0$, the new link reduces the share of migration income retained by household i, raising its migration threshold when background resources are held fixed. The overall effect is thus generally ambiguous and depends on the resources contributed by the new neighbor, which vary with its income, migration decision, and degree.

Therefore, the proposition compares household i's migration before and after forming an additional link, holding other households' migration decisions fixed. However, in equilibrium, the new link can also affect the migration decisions of connected households. Under DARA preferences and partial kinship taxation, these adjustments can feed back into household i's migration incentives through strategic complementarity. Therefore, the overall equilibrium effect of an additional link may differ from its direct effect on the migration threshold.

A related question is how household i's migration incentives change when an existing neighbor $j \in N_i$ forms an additional link with household $k \notin N_i \cup \{i\}$. Unlike the formation

of link ij, this change leaves household i's own degree and migration payoffs unchanged. However, since household j now shares its resources among more connections, household i's background resources decrease by

$$Y_i^{G+jk}(\omega) - Y_i^G(\omega) = -\frac{y_j(\omega) + \alpha e_j(\omega) M_j}{\bar{d}_j(\bar{d}_j + 1)} < 0.$$

Under DARA preferences, a deterministic reduction in background resources raises the migration threshold. However, since the reduction above is generally state-dependent, its effect on the threshold cannot be signed without additional assumptions. Thus, the connectivity of existing neighbors also matters for migration incentives, even when household i's own degree remains unchanged, unlike CARA case. This is implied by the fact that under DARA we have a wealth-effect that the reductions on wealth increases the risk-aversion (i.e. lower migration propensity), but can also change the willingness to undertake an independent migration risk.

Corollary 3.1 (Own Rural Income and the Migration Threshold)**.** *Under the conditions of Proposition 3.5, fix the network G, M_{-i}, and $\mathbf{p}_{-i}$ and consider an increase in household i's rural income by the same amount in every state, $y_i(\omega) \longrightarrow y_i(\omega) + \varepsilon$, $\varepsilon > 0$, $\forall \omega \in \Omega$. Then household i's migration threshold strictly decreases: $\phi_i\big|_{y_i(\omega)+\varepsilon} < \phi_i\big|_{y_i(\omega)}$.*

Thus, under DARA utility, a state-independent increase in household i's own rural income makes migration more attractive. With DARA preferences, an increase in household i's own rural income makes the household more willing to undertake the migration risk. The increase raises background consumption without changing the migration lottery itself. Because absolute risk aversion decreases with consumption under DARA utility, the household becomes better able to absorb the unfavorable migration realization and therefore requires a lower probability of receiving the high remittance outcome. This contrasts directly with the CARA benchmark, under which background income has no effect on the migration threshold.

Below Table 2 summarizes the comparative statics under DARA preferences. For the network comparisons, let ϕ_i^G denote household i's migration threshold under the original network G. Similarly, ϕ_i^{G+ij} denotes its threshold after forming a new link with household $j \notin N_i$, while ϕ_i^{G+jk} denotes its threshold when an existing neighbor $j \in N_i$ forms a new link with household $k \notin N_i \cup \{i\}$.

Table 2: Comparative Statics of the Migration Threshold under DARA

Parameter	**Effect on ϕ_i**	**Interpretation**
High-state return H_i	$\frac{\partial \phi_i}{\partial H_i} < 0$	A higher return in the successful migration state makes migration more attractive and lowers the threshold.
Low-state return L_i	$\frac{\partial \phi_i}{\partial L_i} < 0$	A higher return in the unfavorable migration state reduces potential losses and lowers the threshold.

Parameter	Effect on ϕ_i	Interpretation
Migration cost q	$\frac{\partial \phi_i}{\partial q} > 0$	A higher migration cost reduces net returns in both states and discourages migration.
Background income y_i	$\frac{d\phi_i}{d\varepsilon} < 0$	A uniform increase $y_i(\omega) \to y_i(\omega) + \varepsilon$ raises background resources and, under DARA, encourages risk-taking. Corollary 3.1
Kinship taxation α	Ambiguous	Greater sharing reduces privately retained migration income but increases resources received from migrating neighbors.
Additional link ij	$\phi_i^{G+ij} - \phi_i^G \gtreqless 0$	A new link changes household i's background resources but also reduces the share of its own migration income retained when $\alpha > 0$. The overall effect is ambiguous. Proposition 3.6
Additional link jk	$\phi_i^{G+jk} - \phi_i^G \gtreqless 0$	When neighbor j forms another link, household i receives fewer resources from j, without changing its own migration returns. The effect is ambiguous because the reduction in resources is state-dependent.

Notes: The effects of H_i, L_i, and q hold background resources fixed. The background-income effect refers to a state-independent increase in household i's rural income. The additional-link comparisons hold other households' migration decisions and probabilities fixed. All comparisons assume $L_i < 0 < H_i$ and strictly positive consumption.

4 Conclusion

My framework can also be applied to other risky economic decisions, such as financial investment, entrepreneurship, and agricultural investments, where individuals rely on informal insurance but may also be expected to share the returns from successful projects. The model highlights how these relationships can either encourage or discourage risk-taking, depending on risk preferences and the extent to which returns are shared. More broadly, it illustrates how informal insurance networks can shape economic decisions not only through the resources they provide, but also through the incentives and strategic interactions they create. Importantly, when successful investments by connected households increase others' willingness to take risks, this reinforces investment decisions within the network. Hence, informal insurance arrangements may influence not only who invests, but also how investment decisions spread across connected households.

Finally, my model offers insights into policies aimed at encouraging migration and other risky investments. Providing additional resources or insurance may not have the same effect across households, as their responses depend on risk preferences, network positions, and income-sharing arrangements. Moreover, when migration returns are shared, an intervention targeting one household may also influence the migration incentives of its neighbors. These mechanisms suggest that understanding informal risk-sharing relationships is important when assessing the potential effects of policies designed to encourage productive risk-taking.

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A CARA

A.1 Proof Lemma 3.1

Proof. The migration decision is determined by whether the expected utility gain from migration is positive,

$$\Delta \mathbb{E}^M[u_i] = \mathbb{E}[u_i(M_i = 1)] - \mathbb{E}[u_i(M_i = 0)] \geq 0$$

$$
\begin{aligned}
&\Delta\,\mathbb{E}\{u(c_i(\omega,\mathbf{M}))\} \geq 0 \\
\iff\ &\mathbb{E}\left\{-e^{-r(Y_i(\omega)+e_i-q)}\right\} \geq \mathbb{E}\left\{-e^{-rY_i(\omega)}\right\} \\
\iff\ &\mathbb{E}\left\{e^{-rY_i(\omega)}\right\}\left[-p_i e^{-r(e^H-q)} - (1-p_i)e^{-r(e^L-q)}\right] \geq -\mathbb{E}\left\{e^{-rY_i(\omega)}\right\} \\
\iff\ &-p_i e^{-r(e^H-q)} - (1-p_i)e^{-r(e^L-q)} \geq -1 \\
\iff\ &p_i e^{-r(e^H-q)} + (1-p_i)e^{-r(e^L-q)} \leq 1 \\
\iff\ &p_i\left[e^{-r(e^H-q)} - e^{-r(e^L-q)}\right] \leq 1 - e^{-r(e^L-q)} \\
\iff\ &p_i \geq \frac{e^{-r(e^L-q)}-1}{e^{-r(e^L-q)} - e^{-r(e^H-q)}} \equiv \phi.
\end{aligned}
$$

A.2 Proof Proposition 3.1

Proof. Since migration payoffs do not depend on the migration decisions of other households M_{-i}, each household's optimal decision is independent of its neighbors' actions. Hence, the best-response correspondence of household i is

$$
BR_i(M_{-i}) = \begin{cases} \{1\}, & \text{if } p_i > \phi, \\ \{0\}, & \text{if } p_i < \phi, \end{cases}
$$

If $p_i > \phi$, migration is the unique best response of a household i, independently of the actions of the remaining households. The equilibrium here is unique, but not identical for all households.

If $p_i < \phi$, non-migration is the unique best response of every household. The equilibrium here is unique, but not identical for all households.

A.3 Proof Lemma 3.2

Proof. Migration decision is determined by whether the expected utility gain from migration is positive. Let

$$
Y_i(\omega,\mathbf{M}_{-i}) \equiv y_i(\omega) + \alpha \sum_{j\in N_i} \frac{M_j e_j(\omega)}{\bar{d}_j} + \sum_{j\in N_i} \frac{y_j(\omega)}{\bar{d}_j} - \sum_{j\in N_i} \frac{y_i(\omega)}{\bar{d}_i}.
$$

Then,

$$
\begin{aligned}
&\Delta\,\mathbb{E}^{M}\{u(c_i,\mathbf{M})\} \geq 0 \\
\iff\ &\mathbb{E}\left\{-e^{-r\left[Y_i(\omega,\mathbf{M}_{-i})+\left(1-\alpha+\frac{\alpha}{d_i}\right)e_i(\omega)-q\right]}\right\} \geq \mathbb{E}\left\{-e^{-rY_i(\omega,\mathbf{M}_{-i})}\right\} \\
\iff\ &\mathbb{E}\left\{e^{-rY_i(\omega,\mathbf{M}_{-i})}\right\}\left[-p_i e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^H-q\right]}-(1-p_i)e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^L-q\right]}\right] \\
&\qquad \geq -\mathbb{E}\left\{e^{-rY_i(\omega,\mathbf{M}_{-i})}\right\} \\
\iff\ &-p_i e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^H-q\right]}-(1-p_i)e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^L-q\right]} \geq -1 \\
\iff\ &p_i e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^H-q\right]}+(1-p_i)e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^L-q\right]} \leq 1 \\
\iff\ &p_i\left\{e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^H-q\right]}-e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^L-q\right]}\right\} \leq 1-e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^L-q\right]} \\
\iff\ &p_i \geq \frac{e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^L-q\right]}-1}{e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^L-q\right]}-e^{-r\left[\left(1-\alpha+\frac{\alpha}{d_i}\right)e^H-q\right]}}.
\end{aligned}
$$

A.4 Proof Proposition 3.2

Proof. By Lemma 3.2 and the no-indifference assumption 3.1, household i strictly prefers migration if and only if $p_i > \phi_i$ and non-migration if and only if $p_i < \phi_i$.

Since ϕ_i does not depend on M_{-i}, these comparisons hold for every migration profile of the other households. Define $M_i^* = 1$ if $p_i > \phi_i$ and $M_i^* = 0$ if $p_i < \phi_i$. Every household is choosing its unique best response at M^*, so M^* is a Nash equilibrium.

Moreover, any profile different from M^* assigns some household an action that is not its best response. Thus, the equilibrium is unique.

A.5 Proof for Comparative Statics Table 1

Proof. The migration threshold can then be written as

$$\phi_i = \frac{e^{-rL_i}-1}{e^{-rL_i}-e^{-rH_i}}.$$

Under non-kinship taxation, $\alpha = 0$ and $(1-\alpha) = 1$, so that $H_i = e^H - q$, $L_i = e^L - q$, and the migration threshold is identical across households.

The derivatives with respect to the high-state and low-state migration returns are

$$
\begin{aligned}
\frac{\partial \phi_i}{\partial H_i} &= -\frac{re^{-rH_i}\left(e^{-rL_i}-1\right)}{\left(e^{-rL_i}-e^{-rH_i}\right)^2} < 0, \\
\frac{\partial \phi_i}{\partial L_i} &= -\frac{re^{-rL_i}\left(1-e^{-rH_i}\right)}{\left(e^{-rL_i}-e^{-rH_i}\right)^2} < 0.
\end{aligned}
$$

Since $\frac{\partial H_i}{\partial q} = \frac{\partial L_i}{\partial q} = -1$, the derivative with respect to the migration cost is $\frac{\partial \phi_i}{\partial q} = -\frac{\partial \phi_i}{\partial H_i} - \frac{\partial \phi_i}{\partial L_i} = \frac{r}{e^{-rL_i} - e^{-rH_i}} > 0$.

The derivative with respect to risk aversion is

$$\frac{\partial \phi_i}{\partial r} = \frac{(-L_i)e^{-rL_i}\left(1 - e^{-rH_i}\right) - H_i e^{-rH_i}\left(e^{-rL_i} - 1\right)}{\left(e^{-rL_i} - e^{-rH_i}\right)^2} > 0.$$

The sign follows from $H_i > 0$, $-L_i > 0$, and $\frac{-L_i}{1-e^{rL_i}} > \frac{1}{r} > \frac{H_i}{e^{rH_i}-1}$. Since background income does not enter the expression for ϕ_i, $\frac{\partial \phi_i}{\partial y_i(\omega)} = 0$. Also, $\frac{\partial H_i}{\partial (1/\bar{d}_i)} = \alpha e^H$, $\frac{\partial L_i}{\partial (1/\bar{d}_i)} = \alpha e^L$, we obtain

$$\frac{\partial \phi_i}{\partial (1/\bar{d}_i)} = \alpha e^H \frac{\partial \phi_i}{\partial H_i} + \alpha e^L \frac{\partial \phi_i}{\partial L_i} = -\frac{\alpha r\left[e^H e^{-rH_i}\left(e^{-rL_i} - 1\right) + e^L e^{-rL_i}\left(1 - e^{-rH_i}\right)\right]}{\left(e^{-rL_i} - e^{-rH_i}\right)^2} < 0,$$

provided that $\alpha > 0$ and $e^H, e^L > 0$.

Because $\frac{\partial (1/\bar{d}_i)}{\partial \bar{d}_i} = -\frac{1}{\bar{d}_i^2}$, it follows that

$$\frac{\partial \phi_i}{\partial \bar{d}_i} = -\frac{1}{\bar{d}_i^2}\frac{\partial \phi_i}{\partial (1/\bar{d}_i)} = \frac{\alpha r\left[e^H e^{-rH_i}\left(e^{-rL_i} - 1\right) + e^L e^{-rL_i}\left(1 - e^{-rH_i}\right)\right]}{\bar{d}_i^2\left(e^{-rL_i} - e^{-rH_i}\right)^2} > 0.$$

When migration returns are not shared, $\alpha = 0$, and therefore $\frac{\partial \phi_i}{\partial (1/\bar{d}_i)} = \frac{\partial \phi_i}{\partial \bar{d}_i} = 0$.

B DARA

B.1 Proof Lemma 3.3

Proof. Under $\alpha = 0$, household i's consumption is $c_i(\omega, \mathbf{M}) = Y_i(\omega_i^y) + (e_i(\omega) - q)M_i$. Since migration remittances are privately retained, household i's consumption does not depend on M_{-i}.

The expected utility gain from migration is

$$\begin{aligned}\Delta_i(p_i) = p_i \sum_{\omega_i^y} \Pi_i^y(\omega_i^y)\left[u(Y_i(\omega_i^y) + e^H - q) - u(Y_i(\omega_i^y))\right] \\ + (1 - p_i)\sum_{\omega_i^y} \Pi_i^y(\omega_i^y)\left[u(Y_i(\omega_i^y) + e^L - q) - u(Y_i(\omega_i^y))\right].\end{aligned}$$

Since $e^L - q < 0 < e^H - q$ and u is strictly increasing, the expected utility gain is strictly increasing in p_i. Moreover, $\Delta_i(0) < 0$, $\Delta_i(1) > 0$. Hence, there exists a unique $\phi_i \in (0, 1)$ such that $\Delta_i(\phi_i) = 0$.

Solving this indifference condition gives

$$\phi_i = \frac{\sum_{\omega_i^y} \Pi_i^y(\omega_i^y)\left[u(Y_i(\omega_i^y)) - u(Y_i(\omega_i^y) + e^L - q)\right]}{\sum_{\omega_i^y} \Pi_i^y(\omega_i^y)\left[u(Y_i(\omega_i^y) + e^H - q) - u(Y_i(\omega_i^y) + e^L - q)\right]}.$$

Therefore, household i migrates if $p_i > \phi_i$ and remains if $p_i < \phi_i$. Since $\Delta_i(p_i)$ does not

depend on M_{-i}, migration decisions are independent across households. Thus, the Nash equilibrium is unique whenever $p_i \neq \phi_i$ for every $i \in N$.

B.2 Proof Proposition 3.3

Proof. Take the threshold condition from Lemma 3.3: In particular, households satisfying $p_i > \phi_i$ migrate. However, if $p_i < \phi_i$, household i will choose not to migrate.

Under $p_i = p$: there is a unique Nash Equilibrium whenever $p > \max_{i \in N} \phi_i$ where all households migrate. However, whenever $p < \min_i \phi_i$ there is a unique Nash Equilibrium where no household migrate.

B.3 Proof Proposition 3.4

Proof. Given M_{-i}, household i's consumption can be written as

$$c_i(\omega, \mathbf{M}) = Y_i(\omega, M_{-i}) + \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e_i(\omega) M_i - q M_i.$$

The expected utility gain from migration is

$$\begin{aligned}\Delta_i(p_i; M_{-i}, \mathbf{p}_{-i}) = p_i \sum_{\omega} \Gamma_i(\omega; M_{-i}, \mathbf{p}_{-i}) \left[u(Y_i(\omega, M_{-i}) + H_i) - u(Y_i(\omega, M_{-i}))\right] \\ + (1 - p_i) \sum_{\omega} \Gamma_i(\omega; M_{-i}, \mathbf{p}_{-i}) \left[u(Y_i(\omega, M_{-i}) + L_i) - u(Y_i(\omega, M_{-i}))\right].\end{aligned}$$

Since $L_i < 0 < H_i$ and u is strictly increasing,

$$\Delta_i(0; M_{-i}, \mathbf{p}_{-i}) < 0, \qquad \Delta_i(1; M_{-i}, \mathbf{p}_{-i}) > 0.$$

Moreover, Δ_i is strictly increasing and affine in p_i. Therefore, for every M_{-i} there exists a unique threshold $\phi_i(M_{-i}; \mathbf{p}_{-i}) \in (0, 1)$ such that $\Delta_i(\phi_i; M_{-i}, \mathbf{p}_{-i}) = 0$. Solving this indifference condition yields

$$\phi_i(M_{-i}; \mathbf{p}_{-i}) = \frac{\sum_{\omega} \Gamma_i(\omega; M_{-i}, \mathbf{p}_{-i}) \left[u(Y_i(\omega, M_{-i})) - u(Y_i(\omega, M_{-i}) + L_i)\right]}{\sum_{\omega} \Gamma_i(\omega; M_{-i}, \mathbf{p}_{-i}) \left[u(Y_i(\omega, M_{-i}) + H_i) - u(Y_i(\omega, M_{-i}) + L_i)\right]}.$$

B.4 Proof of Proposition 3.5

Proof. Fix household $i \in N$ and a neighbor $j \in N_i$. Define household i's net migration returns in the high- and low-remittance states as

$$H_i = \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^H - q > 0, \qquad L_i = \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^L - q < 0.$$

Fix $M_{-i,-j}$ and let Y_i collect household i's background resources excluding the migration-

income contribution of household j and household i's own migration return. For a given background-resource random variable Z, define the induced utility $V_Z(x) = \mathbb{E}\left[u(Z+x)\right]$.

I first show that DARA of u implies DARA of the induced utility V_Z. Let $A(c) = -\frac{u''(c)}{u'(c)}$ denote the coefficient of absolute risk aversion of u. Under DARA, $A'(c) < 0$. Since

$$V_Z'(x) = \mathbb{E}[u'(Z+x)], \qquad V_Z''(x) = \mathbb{E}[u''(Z+x)],$$

and, from the definition of absolute risk aversion, $u''(Z+x) = -A(Z+x)u'(Z+x)$, the coefficient of absolute risk aversion of V_Z is

$$A_{V_Z}(x) = -\frac{V_Z''(x)}{V_Z'(x)} = \frac{\mathbb{E}[A(Z+x)u'(Z+x)]}{\mathbb{E}[u'(Z+x)]}.$$

To determine whether V_Z also exhibits DARA, differentiate this expression with respect to x. First,

$$\begin{aligned}\frac{\partial}{\partial x}\left[A(Z+x)u'(Z+x)\right] &= A'(Z+x)u'(Z+x) + A(Z+x)u''(Z+x)\\ &= \left[A'(Z+x) - A(Z+x)^2\right]u'(Z+x),\end{aligned}$$

where the second equality follows from $u''(Z+x) = -A(Z+x)u'(Z+x)$. Moreover,

$$\frac{\partial}{\partial x}\,\mathbb{E}[u'(Z+x)] = \mathbb{E}[u''(Z+x)] = -\,\mathbb{E}[A(Z+x)u'(Z+x)].$$

Applying the quotient rule gives

$$\begin{aligned}A_{V_Z}'(x) &= \frac{\mathbb{E}[(A'(Z+x) - A(Z+x)^2)u'(Z+x)]}{\mathbb{E}[u'(Z+x)]} + \left(\frac{\mathbb{E}[A(Z+x)u'(Z+x)]}{\mathbb{E}[u'(Z+x)]}\right)^2\\ &= \frac{\mathbb{E}[A'(Z+x)u'(Z+x)]}{\mathbb{E}[u'(Z+x)]} - \left[\frac{\mathbb{E}[A(Z+x)^2u'(Z+x)]}{\mathbb{E}[u'(Z+x)]} - \left(\frac{\mathbb{E}[A(Z+x)u'(Z+x)]}{\mathbb{E}[u'(Z+x)]}\right)^2\right].\end{aligned}$$

Under DARA, $A'(Z+x) < 0$, and since $u'(Z+x) > 0$, $\frac{\mathbb{E}[A'(Z+x)u'(Z+x)]}{\mathbb{E}[u'(Z+x)]} < 0$. The expression in square brackets is a variance under weights proportional to marginal utility $u'(Z+x)$ and is therefore non-negative. Hence, $A_{V_Z}'(x) < 0$. Thus, the induced utility V_Z also exhibits DARA.

I now establish how a deterministic increase in background resources affects the migration threshold. Let $t \geq 0$. Conditional on background resources $Z+t$, the expected utility gain from migration is

$$\Delta_i(p_i; t) = p_i\left[V_Z(t+H_i) - V_Z(t)\right] + (1-p_i)\left[V_Z(t+L_i) - V_Z(t)\right].$$

The migration threshold $\phi_i(t)$ is defined by $\Delta_i(\phi_i(t); t) = 0$. Therefore,

$$\phi_i(t)\left[V_Z(t+H_i) - V_Z(t)\right] = (1-\phi_i(t))\left[V_Z(t) - V_Z(t+L_i)\right],$$

which implies $\frac{\phi_i(t)}{1-\phi_i(t)} = R_i(t) \equiv \frac{V_Z(t)-V_Z(t+L_i)}{V_Z(t+H_i)-V_Z(t)}$.

Using the fundamental theorem of calculus,

$$V_Z(t) - V_Z(t+L_i) = \int_{L_i}^{0} V_Z'(t+s)\,ds, \qquad V_Z(t+H_i) - V_Z(t) = \int_{0}^{H_i} V_Z'(t+s)\,ds,$$

so that $R_i(t) = \dfrac{\int_{L_i}^{0} V_Z'(t+s)\,ds}{\int_{0}^{H_i} V_Z'(t+s)\,ds}$. To determine how $R_i(t)$ changes with t, divide both the numerator and denominator by $V_Z'(t) > 0$:

$$R_i(t) = \frac{\int_{L_i}^{0} \dfrac{V_Z'(t+s)}{V_Z'(t)}\,ds}{\int_{0}^{H_i} \dfrac{V_Z'(t+s)}{V_Z'(t)}\,ds}.$$

For $s \in [L_i, 0)$,

$$\frac{\partial}{\partial t}\log\left(\frac{V_Z'(t+s)}{V_Z'(t)}\right) = \frac{V_Z''(t+s)}{V_Z'(t+s)} - \frac{V_Z''(t)}{V_Z'(t)} = -A_{V_Z}(t+s) + A_{V_Z}(t) < 0,$$

because $t+s < t$ and A_{V_Z} is decreasing. Therefore, $\frac{V_Z'(t+s)}{V_Z'(t)}$ is strictly decreasing in t for every $s \in [L_i, 0)$, implying that $\int_{L_i}^{0} \frac{V_Z'(t+s)}{V_Z'(t)}\,ds$ is strictly decreasing in t.

Similarly, for $s \in (0, H_i]$, $\frac{\partial}{\partial t}\log\left(\frac{V_Z'(t+s)}{V_Z'(t)}\right) = -A_{V_Z}(t+s) + A_{V_Z}(t) > 0$, because $t+s > t$ and A_{V_Z} is decreasing. Therefore, $\frac{V_Z'(t+s)}{V_Z'(t)}$ is strictly increasing in t for every $s \in (0, H_i]$, implying that $\int_{0}^{H_i} \frac{V_Z'(t+s)}{V_Z'(t)}\,ds$ is strictly increasing in t.

Hence, the numerator of $R_i(t)$ after normalization is strictly decreasing in t, while the denominator is strictly increasing in t. It follows that $R_i'(t) < 0$. Since $\phi_i(t) = \frac{R_i(t)}{1+R_i(t)}$ and $R/(1+R)$ is strictly increasing in R, it follows that $\phi_i'(t) < 0$. Thus, under DARA, a deterministic positive increase in background resources lowers household i's migration threshold.

I now consider household j's migration. If $M_j = 0$, household j contributes no migration income to household i. If $M_j = 1$, household j contributes $t_j = \frac{\alpha e_j}{\bar{d}_j}$. Since $e^L > 0$ and $\alpha > 0$, $t_j > 0$ in every state. In particular,

$$t_j = \begin{cases} \dfrac{\alpha e^H}{\bar{d}_j}, & \text{with probability } p_j, \\ \dfrac{\alpha e^L}{\bar{d}_j}, & \text{with probability } 1-p_j. \end{cases}$$

For a given realization of t_j, define $A_i(t_j) = V_Z(t_j) - V_Z(t_j + L_i)$, $B_i(t_j) = V_Z(t_j + H_i) - V_Z(t_j)$. Since $L_i < 0 < H_i$ and V_Z is strictly increasing, $A_i(t_j) > 0, B_i(t_j) > 0$.

When $M_j = 0$, $t_j = 0$, and the migration threshold satisfies $\frac{\phi_i^0}{1-\phi_i^0} = \frac{A_i(0)}{B_i(0)}$. From the

argument above, $R_i(t) = \frac{A_i(t)}{B_i(t)}$ is strictly decreasing in t. Since $t_j > 0$ for both realizations of e_j, $\frac{A_i(t_j)}{B_i(t_j)} < \frac{A_i(0)}{B_i(0)}$ state by state. Since all terms are strictly positive, this is equivalent to $A_i(t_j)B_i(0) < A_i(0)B_i(t_j)$. Taking expectations over household j's migration-return shock preserves the inequality: $\mathbb{E}_{e_j}[A_i(t_j)]B_i(0) < A_i(0)\,\mathbb{E}_{e_j}[B_i(t_j)]$. Dividing by the strictly positive term $\mathbb{E}_{e_j}[B_i(t_j)]B_i(0)$ gives $\frac{\mathbb{E}_{e_j}[A_i(t_j)]}{\mathbb{E}_{e_j}[B_i(t_j)]} < \frac{A_i(0)}{B_i(0)}$.

When $M_j = 1$, the migration threshold ϕ_i^1 is characterized by $\frac{\phi_i^1}{1-\phi_i^1} = \frac{\mathbb{E}_{e_j}[A_i(t_j)]}{\mathbb{E}_{e_j}[B_i(t_j)]}$. Using the two possible realizations of e_j,

$$\frac{\phi_i^1}{1-\phi_i^1} = \frac{p_j A_i\left(\frac{\alpha e^H}{\bar{d}_j}\right) + (1-p_j)A_i\left(\frac{\alpha e^L}{\bar{d}_j}\right)}{p_j B_i\left(\frac{\alpha e^H}{\bar{d}_j}\right) + (1-p_j)B_i\left(\frac{\alpha e^L}{\bar{d}_j}\right)}.$$

Therefore, $\frac{\phi_i^1}{1-\phi_i^1} < \frac{\phi_i^0}{1-\phi_i^0}$. Since $\phi/(1-\phi)$ is strictly increasing in ϕ for $\phi \in (0,1)$, $\phi_i^1 < \phi_i^0$. Equivalently,

$$\phi_i\left(M_j = 1, M_{-i,-j}; \mathbf{p}_{-i}\right) < \phi_i\left(M_j = 0, M_{-i,-j}; \mathbf{p}_{-i}\right).$$

Hence, migration by household j lowers the migration threshold of household i. Equivalently, household i's best response is non-decreasing in M_j, so migration decisions are strategic complements along existing risk-sharing links. The log-utility case is a particular case of this result.

Importantly, this mechanism requires both remittance sharing and decreasing absolute risk aversion. If remittances were private, household j's migration would not affect household i's resources. If preferences were CARA, the increase in background resources would factor out of household i's migration comparison. Strategic complementarity therefore arises from the interaction between the network transfer generated by migration and the dependence of risk-taking incentives on background consumption.

Hence, household i migrates when $p_i > \phi_i(M_{-i}; \mathbf{p}_{-i})$ and remains when $p_i < \phi_i(M_{-i}; \mathbf{p}_{-i})$. By Proposition 3.5, under strict DARA and partial kinship taxation, the game exhibits strategic complementarities: each household's best-response correspondence is non-decreasing in the migration decisions of the other households. Since the set of migration profiles $\{0,1\}^n$ is a finite complete lattice, the existence of increasing best responses implies, by Tarski's fixed-point theorem, that at least one pure-strategy Nash equilibrium exists. Moreover, the set of pure-strategy Nash equilibria has a smallest and a largest element under the component wise order. However, strategic complementarities do not imply uniqueness, so multiple equilibria may arise.

B.5 Proof Proposition 3.6

Proof. Forming link ij increases the closed degrees of households i and j from $\bar{d}_i$ and $\bar{d}_j$ to $\bar{d}_i + 1$ and $\bar{d}_j + 1$. Therefore,

$$Y_i^{G+ij}(\omega) = \frac{y_i(\omega)}{\bar{d}_i + 1} + \sum_{k \in N_i} \left[\frac{y_k(\omega)}{\bar{d}_k} + \alpha \frac{e_k(\omega) M_k}{\bar{d}_k} \right] + \frac{y_j(\omega) + \alpha e_j(\omega) M_j}{\bar{d}_j + 1}.$$

Subtracting $Y_i^G(\omega)$ gives

$$Y_i^{G+ij}(\omega) - Y_i^G(\omega) = y_i(\omega) \left(\frac{1}{\bar{d}_i + 1} - \frac{1}{\bar{d}_i} \right) + \frac{y_j(\omega) + \alpha e_j(\omega) M_j}{\bar{d}_j + 1}.$$

Since $\frac{1}{d_i+1} - \frac{1}{d_i} = -\frac{1}{d_i(d_i+1)}$, we obtain

$$Y_i^{G+ij}(\omega) - Y_i^G(\omega) = \frac{y_j(\omega) + \alpha e_j(\omega) M_j}{\bar{d}_j + 1} - \frac{y_i(\omega)}{\bar{d}_i(\bar{d}_i + 1)}.$$

The first term is the additional resources received from the new neighbor j, while the second term is the reduction in the fraction of household i's own rural income that it retains after its degree increases. Since these two effects have opposite signs, the effect of the additional link on Y_i is generally ambiguous and may depend on the realization of the shocks.

The additional link also changes the fraction of household i's own migration income that it retains. For either $e^s \in \{e^H, e^L\}$,

$$\left(1 - \alpha + \frac{\alpha}{\bar{d}_i + 1}\right) e^s - q - \left[\left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^s - q \right] = -\frac{\alpha e^s}{\bar{d}_i(\bar{d}_i + 1)} < 0.$$

Thus, for $\alpha > 0$, holding household i's background resources fixed, its consumption under migration is lower after the additional link for both realizations of its own migration return. Its consumption under non-migration is unchanged. Hence the expected utility gain from migration falls for every p_i. Since the migration gain is strictly increasing in p_i, the migration threshold rises when only this effect is considered.

The total effect must also account for the change from Y_i^G to Y_i^{G+ij}. Under network G, household i is indifferent at $p_i = \phi_i^G$, so

$$\mathbb{E}\left[\phi_i^G u\left(Y_i^G + \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^H - q \right) + (1 - \phi_i^G) u\left(Y_i^G + \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right) e^L - q \right) - u(Y_i^G) \right] = 0.$$

Under $G+ij$, the expected utility gain from migration is strictly increasing in p_i, because

$$\frac{\partial \Delta \mathbb{E}_i^{M,G+ij}}{\partial p_i} = \mathbb{E}\Bigg[u\left(Y_i^{G+ij} + \left(1 - \alpha + \frac{\alpha}{\bar{d}_i + 1}\right) e^H - q \right) - u\left(Y_i^{G+ij} + \left(1 - \alpha + \frac{\alpha}{\bar{d}_i + 1}\right) e^L - q \right) \Bigg] > 0.$$

The inequality follows because $e^H > e^L$ and consumption is strictly positive. Therefore, to determine whether the new threshold is above or below the old threshold, it is sufficient to evaluate the migration gain under $G + ij$ at $p_i = \phi_i^G$. At the indifference point $\Delta\,\mathbb{E}_i^{M,G+ij}\big|_{p_i=\phi_i^G} = 0$ holds.

If that gain is positive, household i already prefers migration under $G + ij$ at the probability at which it was indifferent under G. Since the migration gain is increasing in p_i, the new threshold must therefore be lower: $\Delta\,\mathbb{E}_i^{M,G+ij}\big|_{p_i=\phi_i^G} > 0 \quad \Longrightarrow \quad \phi_i^{G+ij} < \phi_i^G$.

If instead $\Delta\,\mathbb{E}_i^{M,G+ij}\big|_{p_i=\phi_i^G} < 0$, the new threshold must be higher, while equality implies that the two thresholds coincide.

To obtain the explicit condition, evaluate the migration gain under $G + ij$ at $p_i = \phi_i^G$:

$$\mathbb{E}\left[\phi_i^G u\left(Y_i^{G+ij} + \left(1-\alpha+\frac{\alpha}{\bar{d}_i+1}\right)e^H - q\right) + (1-\phi_i^G)u\left(Y_i^{G+ij} + \left(1-\alpha+\frac{\alpha}{\bar{d}_i+1}\right)e^L - q\right) - u(Y_i^{G+ij})\right].$$

Subtracting the indifference condition under G from the migration gain under $G + ij$, we obtain

$$\begin{aligned}\mathbb{E}\Bigg[&\phi_i^G\left(u\left(Y_i^{G+ij} + \left(1-\alpha+\frac{\alpha}{\bar{d}_i+1}\right)e^H - q\right) - u\left(Y_i^{G} + \left(1-\alpha+\frac{\alpha}{\bar{d}_i}\right)e^H - q\right)\right)\\ &+ (1-\phi_i^G)\left(u\left(Y_i^{G+ij} + \left(1-\alpha+\frac{\alpha}{\bar{d}_i+1}\right)e^L - q\right) - u\left(Y_i^{G} + \left(1-\alpha+\frac{\alpha}{\bar{d}_i}\right)e^L - q\right)\right)\\ &- u(Y_i^{G+ij}) + u(Y_i^G)\Bigg]\end{aligned}$$

A positive value implies $\phi_i^{G+ij} < \phi_i^G$, a negative value implies $\phi_i^{G+ij} > \phi_i^G$, while equality implies $\phi_i^{G+ij} = \phi_i^G$.

Thus, the additional link generates two effects. It changes household i's background resources, which may increase or decrease depending on the resources contributed by the new neighbor relative to the reduction in household i's retained rural income. The background-resource effect is more favorable when household j contributes more resources to i, for instance when $y_j(\omega)$ is higher, when $M_j = 1$ and $e_j(\omega)$ is higher, or when $\bar{d}_j$ is smaller so that a larger share of j's resources reaches household i. Moreover, the reduction in household i's retained rural income becomes smaller in magnitude as $\bar{d}_i$ increases.

At the same time, for $\alpha > 0$, the increase in $\bar{d}_i$ unambiguously reduces the fraction of household i's own migration income that it retains. This effect is stronger when α is larger and when the realization of household i's migration income is larger, while its magnitude becomes smaller as $\bar{d}_i$ increases. Hence, conditional on the background-resource effect, stronger kinship taxation raises the upward pressure of the new link on the migration threshold.

Consequently, a link with a relatively resource-rich or less-connected household j, particularly when j migrates and contributes remittances, is more likely to generate a

positive background-resource effect that offsets the reduction in household i's retained migration income. Conversely, when the resources received from j are small relative to the dilution of household i's own resources and migration income, the additional link tends to raise the migration threshold. The total effect on the migration threshold is therefore generally ambiguous.

B.6 Proof Corollary 3.1

Proof. Recall that household i's resources excluding its own migration return and under kinship taxation of migration are

$$Y_i(\omega, M_{-i}) = \frac{y_i(\omega)}{\bar{d}_i} + \sum_{j \in N_i} \left[\frac{y_j(\omega)}{\bar{d}_j} + \alpha \frac{e_j(\omega) M_j}{\bar{d}_j} \right].$$

Therefore, increasing $y_i(\omega)$ by $\varepsilon > 0$ in every state changes household i's resources according to $Y_i(\omega, M_{-i}) \longrightarrow Y_i(\omega, M_{-i}) + \frac{\varepsilon}{\bar{d}_i}$. Since $\bar{d}_i > 0$, $\frac{\varepsilon}{\bar{d}_i} > 0$.

The proof of Proposition 3.5 establishes that adding the same strictly positive amount to $Y_i(\omega, M_{-i})$ in every state lowers the threshold odds $\frac{\phi_i(M_{-i};\mathbf{p}_{-i})}{1-\phi_i(M_{-i};\mathbf{p}_{-i})}$. Indeed, the positive increase in background resources reduces the expected utility loss associated with the low migration realization proportionally more than it reduces the expected utility gain associated with the high migration realization. Therefore, $\frac{\phi_i}{1-\phi_i}\Big|_{y_i(\omega)+\varepsilon} < \frac{\phi_i}{1-\phi_i}\Big|_{y_i(\omega)}$. Since $\frac{\partial}{\partial \phi_i}\left(\frac{\phi_i}{1-\phi_i}\right) = \frac{1}{(1-\phi_i)^2} > 0$, it follows that $\phi_i\big|_{y_i(\omega)+\varepsilon} < \phi_i\big|_{y_i(\omega)}$.

B.7 Proof Comparative Statics Table 2

Proof. Fix the network G, the migration choices of other households, and their success probabilities. Recall that $\phi_i = \frac{A_i}{A_i+B_i}$, where $A_i = \mathbb{E}[u(Y_i) - u(Y_i + L_i)] > 0$ and $B_i = \mathbb{E}[u(Y_i + H_i) - u(Y_i)] > 0$, with $L_i < 0 < H_i$. For any parameter x,

$$\frac{\partial \phi_i}{\partial x} = \frac{B_i\, \partial A_i/\partial x - A_i\, \partial B_i/\partial x}{(A_i + B_i)^2}.$$

High-state return. Holding background resources and L_i fixed, we have $\partial A_i/\partial H_i = 0$ and $\partial B_i/\partial H_i = \mathbb{E}[u'(Y_i + H_i)] > 0$. Consequently,

$$\frac{\partial \phi_i}{\partial H_i} = -\frac{A_i\, \mathbb{E}[u'(Y_i + H_i)]}{(A_i + B_i)^2} < 0.$$

Low-state return. Similarly, $\partial B_i/\partial L_i = 0$ and $\partial A_i/\partial L_i = -\,\mathbb{E}[u'(Y_i + L_i)] < 0$. Therefore,

$$\frac{\partial \phi_i}{\partial L_i} = -\frac{B_i\, \mathbb{E}[u'(Y_i + L_i)]}{(A_i + B_i)^2} < 0.$$

Migration cost. Since $\partial L_i/\partial q = \partial H_i/\partial q = -1$, we obtain $\partial A_i/\partial q = \mathbb{E}[u'(Y_i + L_i)] > 0$ and

$\partial B_i/\partial q = -\mathbb{E}[u'(Y_i + H_i)] < 0$. Thus,

$$\frac{\partial \phi_i}{\partial q} = \frac{B_i \, \mathbb{E}[u'(Y_i + L_i)] + A_i \, \mathbb{E}[u'(Y_i + H_i)]}{(A_i + B_i)^2} > 0.$$

Background income Follow Corollary 3.1 proof.

Kinship taxation. An increase in α affects both background resources and the retained migration returns. In particular, $\partial Y_i/\partial \alpha = \sum_{j \in N_i} e_j M_j / \bar{d}_j$, while $\partial H_i/\partial \alpha = -(1 - 1/\bar{d}_i)e^H$ and $\partial L_i/\partial \alpha = -(1 - 1/\bar{d}_i)e^L$.

Writing $Y_\alpha = \partial Y_i/\partial \alpha$, $H_\alpha = \partial H_i/\partial \alpha$, and $L_\alpha = \partial L_i/\partial \alpha$, we have

$$\frac{\partial A_i}{\partial \alpha} = \mathbb{E}[u'(Y_i)Y_\alpha - u'(Y_i + L_i)(Y_\alpha + L_\alpha)],$$
$$\frac{\partial B_i}{\partial \alpha} = \mathbb{E}[u'(Y_i + H_i)(Y_\alpha + H_\alpha) - u'(Y_i)Y_\alpha].$$

The direct reduction in retained migration returns raises the threshold, holding background resources fixed. However, changes in remittances received from migrating neighbors also affect background resources. The resulting derivative is therefore not generally signed by the maintained assumptions.

Additional link. Consider the formation of link ij, where $j \notin N_i$. As established in Proposition 3.6,

$$Y_i^{G+ij} - Y_i^G = \frac{y_j + \alpha e_j M_j}{\bar{d}_j + 1} - \frac{y_i}{\bar{d}_i(\bar{d}_i + 1)}.$$

The link also reduces the retained migration returns when $\alpha > 0$. Define A_i^G, B_i^G and A_i^{G+ij}, B_i^{G+ij} using the corresponding background resources and migration payoffs. Then

$$\phi_i^{G+ij} - \phi_i^G = \frac{A_i^{G+ij} B_i^G - A_i^G B_i^{G+ij}}{(A_i^{G+ij} + B_i^{G+ij})(A_i^G + B_i^G)}.$$

Since the denominator is positive, the sign is determined exactly by $A_i^{G+ij} B_i^G - A_i^G B_i^{G+ij}$. The maintained assumptions do not impose a general sign on this expression.

C Extensions: Random Network and Cost Sharing

C.1 Random Network

I now consider an extension in which the network is randomly generated before households make their migration decisions. Once realized, the network is observed by all households, and migration decisions follow the baseline model.

Suppose G is drawn from an Erdős–Rényi random graph, $G \sim \mathcal{G}(n, \rho)$, where $\rho \in [0, 1]$ is the probability that any two distinct households are connected. Links are formed independently, so household i's degree satisfies

$$d_i \sim \text{Binomial}(n - 1, \rho), \qquad \mathbb{E}_G[\bar{d}_i] = 1 + (n - 1)\rho.$$

A higher ρ therefore corresponds to a more densely connected network.

Proposition C.1 (Network Density and Migration under CARA)**.** *Suppose households have CARA preferences and migration income is partially shared, with* $0 < \alpha < 1$*. Assume that* $L_i(d) < 0 < H_i(d)$ *for every feasible degree* d*.*

For any $0 \leq \rho_1 < \rho_2 \leq 1$*,* $\mathbb{E}_G[\phi_i(d_i) \mid \rho_2] > \mathbb{E}_G[\phi_i(d_i) \mid \rho_1]$*, provided* $n \geq 2$*.*

Proof. Under CARA preferences, the migration threshold depends only on household i's degree and is strictly increasing in d_i when migration income is partially shared. Since $d_i \sim \text{Binomial}(n-1, \rho)$, the expected threshold is

$$\mathbb{E}_G[\phi_i(d_i) \mid \rho] = \sum_{d=0}^{n-1} \binom{n-1}{d} \rho^d (1-\rho)^{n-1-d} \phi_i(d).$$

For $0 < \rho < 1$, differentiating with respect to ρ gives

$$\frac{\partial}{\partial \rho} \mathbb{E}_G[\phi_i(d_i) \mid \rho] = \sum_{d=0}^{n-1} \phi_i(d) \binom{n-1}{d} \Big[d\rho^{d-1}(1-\rho)^{n-1-d} - (n-1-d)\rho^d (1-\rho)^{n-2-d} \Big].$$

Using the binomial identities $d\binom{n-1}{d} = (n-1)\binom{n-2}{d-1}$, $(n-1-d)\binom{n-1}{d} = (n-1)\binom{n-2}{d}$, and reindexing the first sum, we obtain

$$\frac{\partial}{\partial \rho} \mathbb{E}_G[\phi_i(d_i) \mid \rho] = (n-1) \sum_{d=0}^{n-2} \binom{n-2}{d} \rho^d (1-\rho)^{n-2-d} \times [\phi_i(d+1) - \phi_i(d)].$$

Since $\phi_i(d+1) > \phi_i(d)$ under CARA and partial kinship taxation, every term in the sum is nonnegative, and the sum is strictly positive for $n \geq 2$. Hence, $\frac{\partial}{\partial \rho} \mathbb{E}_G[\phi_i(d_i) \mid \rho] > 0$. Continuity at $\rho = 0$ and $\rho = 1$ then implies the stated strict inequality for any $\rho_2 > \rho_1$.

Under DARA preferences, the effect of network density is not generally signed by the preceding argument. Unlike under CARA, the migration threshold depends on the household's background resources, which vary with the realized network and its neighbors' migration decisions. Additional connections change both the resources received through the network and the fraction of migration income retained by the household. When migration income is shared, changes in network structure may also affect strategic interactions among households.

Consequently, the degree-based monotonicity used in the CARA proof is insufficient to establish the sign of the expected threshold under DARA. Its response to network density depends on the changes in background resources, risk exposure, and sharing obligations.

C.2 Shared-Cost

Now think of the case where migrants will not only transfer part of the remittances to their peers, but also part of the migration cost q. Now then it is not obvious than having peer migrants will increase household i's consumption.

Proposition C.2. *Suppose households have CARA preferences, with $0 < \alpha < 1$, and migration income and migration costs are shared at the same rate α.*

Fix household $i \in N$, a neighbor $j \in N_i$, and $M_{-i,-j}$. Define

$$H_i = \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right)(e^H - q) > 0, \qquad L_i = \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right)(e^L - q) < 0,$$

The equilibrium becomes

$$M_i^* = \begin{cases} 1, & \text{if } p_i > \phi_i, \\ 0, & \text{if } p_i < \phi_i. \end{cases}$$

Also, under this case ϕ_i is smaller than under no kinship taxation on migration cost. Then, migration now is easier to reach by any household i since they retain more gains from migration due to q sharing.

Proof. Under CARA preferences, household i prefers migration if

$$p_i \mathbb{E}\left[-e^{-r(Y_i+H_i)}\right] + (1-p_i)\mathbb{E}\left[-e^{-r(Y_i+L_i)}\right] > \mathbb{E}\left[-e^{-rY_i}\right].$$

Since migration returns are independent of rural income shocks, the common term $\mathbb{E}[e^{-rY_i}]$ factors out. Therefore, the condition becomes

$$p_i e^{-rH_i} + (1-p_i)e^{-rL_i} < 1.$$

Rearranging gives

$$p_i > \frac{e^{-rL_i} - 1}{e^{-rL_i} - e^{-rH_i}} \equiv \phi_i,$$

where

$$H_i = \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right)(e^H - q), \qquad L_i = \left(1 - \alpha + \frac{\alpha}{\bar{d}_i}\right)(e^L - q).$$

Since $H_i > 0 > L_i$, we have $\phi_i \in (0, 1)$. The threshold does not depend on the migration decisions of other households. Hence, provided $p_i \neq \phi_i$ for every household, each household has a unique optimal migration decision, yielding a unique equilibrium.

Proposition C.3 (Strategic Interaction with Shared Migration Costs)**.** *Suppose households have DARA preferences, with $0 < \alpha < 1$, and strictly positive consumption in every state. Migration income and migration costs are shared at the same rate α.*

By using H_i and L_i above, and $t_j^H = \frac{\alpha(e^H - q)}{\bar{d}_j} > 0$, $t_j^L = \frac{\alpha(e^L - q)}{\bar{d}_j} < 0$.

Let Z denote household i's background resources excluding household j's migration contribution, and define $V_Z(x) = \mathbb{E}[u(Z + x)]$.

For $s \in \{0, t_j^L, t_j^H\}$, define

$$A_i(s) = V_Z(s) - V_Z(s + L_i) > 0, \qquad B_i(s) = V_Z(s + H_i) - V_Z(s) > 0.$$

Then there exists a unique $p_j^* \in (0,1)$ such that the indifference condition simplifies to

$$p_j^* = \frac{A_i(t_j^L)B_i(0) - A_i(0)B_i(t_j^L)}{\left[A_i(t_j^L)B_i(0) - A_i(0)B_i(t_j^L)\right] - \left[A_i(t_j^H)B_i(0) - A_i(0)B_i(t_j^H)\right]}.$$

Moreover,

$$\phi_i(M_j = 1, M_{-i,-j}; \mathbf{p}_{-i}) \begin{cases} < \phi_i(M_j = 0, M_{-i,-j}; \mathbf{p}_{-i}), & \text{if } p_j > p_j^*, \\ = \phi_i(M_j = 0, M_{-i,-j}; \mathbf{p}_{-i}), & \text{if } p_j = p_j^*, \\ > \phi_i(M_j = 0, M_{-i,-j}; \mathbf{p}_{-i}), & \text{if } p_j < p_j^*. \end{cases}$$

Thus, migration decisions are strategic complements when $p_j > p_j^*$ and strategic substitutes when $p_j < p_j^*$.

Proof. Under DARA, the induced utility V_Z also exhibits DARA, as established in the proof of Proposition 3.5. Therefore, $\frac{A_i(s)}{B_i(s)}$ is strictly decreasing in s.

Since $t_j^L < 0 < t_j^H$, it follows that $\frac{A_i(t_j^H)}{B_i(t_j^H)} < \frac{A_i(0)}{B_i(0)} < \frac{A_i(t_j^L)}{B_i(t_j^L)}$.

All terms are strictly positive, so $A_i(t_j^H)B_i(0) - A_i(0)B_i(t_j^H) < 0$, and $A_i(t_j^L)B_i(0) - A_i(0)B_i(t_j^L) > 0$. When $M_j = 0$, the migration threshold satisfies $\frac{\phi_i^0}{1-\phi_i^0} = \frac{A_i(0)}{B_i(0)}$. When $M_j = 1$, the migration threshold satisfies

$$\frac{\phi_i^1}{1-\phi_i^1} = \frac{p_j A_i(t_j^H) + (1-p_j)A_i(t_j^L)}{p_j B_i(t_j^H) + (1-p_j)B_i(t_j^L)}.$$

Since all denominators are strictly positive, $\phi_i^1 < \phi_i^0$ if and only if

$$\left[p_j A_i(t_j^H) + (1-p_j)A_i(t_j^L)\right] B_i(0) < A_i(0)\left[p_j B_i(t_j^H) + (1-p_j)B_i(t_j^L)\right].$$

Rearranging,

$$p_j\left[A_i(t_j^H)B_i(0) - A_i(0)B_i(t_j^H)\right] + (1-p_j)\left[A_i(t_j^L)B_i(0) - A_i(0)B_i(t_j^L)\right] < 0.$$

The first bracket is strictly negative, while the second is strictly positive.[2] Hence, the left-hand side is strictly decreasing and affine in p_j, positive at $p_j = 0$ and negative at $p_j = 1$. It therefore crosses zero exactly once (when $\phi^0 = \phi^1$) and the indifference condition

[2] Under DARA, the ratio $A_i(t)/B_i(t)$ is strictly decreasing in t. Since $t_j^L < 0 < t_j^H$, we have $\frac{A_i(t_j^H)}{B_i(t_j^H)} < \frac{A_i(0)}{B_i(0)} < \frac{A_i(t_j^L)}{B_i(t_j^L)}$. As $B_i(t) > 0$, cross-multiplication gives the stated signs.

is reached at,

$$p_j^* = \frac{A_i(t_j^L)B_i(0) - A_i(0)B_i(t_j^L)}{\Big[A_i(t_j^L)B_i(0) - A_i(0)B_i(t_j^L)\Big] - \Big[A_i(t_j^H)B_i(0) - A_i(0)B_i(t_j^H)\Big]}.$$

Consequently, $\phi_i^1 < \phi_i^0 \iff p_j > p_j^*$, $\phi_i^1 > \phi_i^0 \iff p_j < p_j^*$.

At $p_j = p_j^*$, the two thresholds coincide. Thus, household j's migration encourages household i to migrate when $p_j > p_j^*$ and discourages it when $p_j < p_j^*$.